

The Celtics Live Win Probability Model (CLWP): Building and Testing a Team-Specific Real-Time Win Probability Model on Eight Seasons of Play-by-Play Data

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Abstract

This study involved building and testing the Celtics Live Win Probability model, referred to throughout as the CLWP model. CLWP is a real-time win probability model calibrated specifically to the Boston Celtics and trained on 308,975 play-by-play events across 636 regular season games from the 2016-17 season through the 2023-24 season. The research question asked is whether a model that knows which team is playing, which players are on the floor, and how the game is proceeding can predict the outcome better than a model that uses only the score and the clock.

The CLWP model was evaluated using the leave one season out cross validation, so that every prediction reported here comes from a version of the model that had never seen that season. Out of fold, the CLWP model achieves a Brier score of 0.1630, an AUC of 0.8155 and a Brier skill of 28.8 percent over the base rate. The comparison model, a transparent reimplement of the standard margin and clock approach and referred to throughout as the generic model based on ESPN, achieves a Brier score of 0.1641.

Pooled across all events, the two models are not distinguishable. The difference is 0.0011 Brier, in favor of the CLWP model, with a 95 percent cluster bootstrap interval of $[-0.0029, +0.0048]$, which contains zero. Read on its own, that pooled figure would suggest the team-specific information adds nothing. The phase by phase results show that this reading is incomplete. The CLWP model is better calibrated than the generic model based on ESPN in seven of the eight game phases examined, and its advantage is largest exactly where a live win probability display is hardest to get right, which is early in the game before the score has separated. In the first six minutes the generic model based on ESPN has negative skill, meaning it does worse than simply predicting Boston's long run win rate, while the CLWP model manages positive skill.

The study also identifies three key findings which are the errors in the NBA's historical play-by-play data, a feature memorization problem in machine learning models, and a limitation that affects all in-game win probability models before tip-off.

1. Introduction

Win probability information is now a standard part of how basketball is broadcast and discussed. A win probability value appears on screen, it updates after every play, and viewers treat it as a statement about the game in front of them. The models behind those numbers are usually generic. They are fitted on every team at once and they take a small number of inputs, most commonly the score margin, the time remaining, and the interaction between the two.

The question this project set out to answer is whether that generality of these models results in poor probability performance? Is there a cost that is paid as a result of using the generic models?. If a

model were built for one team, and given information specific to that team, would it produce more accurate probabilities for that team's games than a generic model does?

The Boston Celtics are a reasonable subject for that question. Over the eight seasons studied here, the team was consistently competitive without being uniformly dominant. The Celtics finished 413-223 across 636 regular season games having season win totals ranging from 36 to 64. That variation matters, since a team that wins almost every game or loses almost every game produces a model resulting in statistical insignificance. Basically, we will learn very little by studying such teams.

The three questions below and their associated answers produce the body and organization of this work.

1. Does a Celtics-specific model that uses game state information predict Boston's games better than a generic ESPN based model that uses only the score and the clock?
2. Does adding information from which five players are currently on the floor playing improve the prediction further?
3. Does adding information about the quality of the opponent improve the prediction further?

The first question is the one the project was originated to answer. The second and third are additions, and both additional questions were tested with the same standard applied to the first.

The contributions this study makes are as follows: It delivers a validated, reproducible pipeline that turns raw NBA API responses into a modeling dataset with every intermediate step independently checked. It reports the model's performance honestly, including the places where the team-specific model does not surpass the simpler one. It documents real defects in the historical play-by-play feed that would silently corrupt any study using that feed without checking it. The model identifies a cardinality artifact that makes standard feature importance reasoning unreliable in event-level sports models and produces a working interactive dashboard in which every displayed probability is out of fold.

2. Background and Related Work

The standard in-game win probability model, in NBA basketball, is a logistic regression on score margin and time remaining, with an interaction term between the two. Comparable work in other sports has used more flexible learners on the same kind of state, for example the NFL uses the random forest approach of Lock and Nettleton (2014). The interaction between time and score is what makes the model work, since a five point lead means something very different with two minutes left than it does with thirty minutes left.

The registration form for this directed research specified a benchmark against ESPN's win probability model. That comparison is not possible. ESPN does not publish their model, so no comparison is possible. What stands in its place is a transparent reimplement of the standard margin and clock approach of the kind ESPN is understood to use. It is referred to throughout this paper as the generic model. Naming it this way is intended to be precise about what it is: a faithful implementation of the published approach, not a copy of any proprietary system. A unit test in the codebase asserts that no model tier name in the pipeline contains the string "espn", so the distinction is not lost.

The model constructed for this study is also informed by two related areas of research. The first is the player valuation literature, which adjusts plus or minus and its evolving versions attempt to separate a player's contribution from that of his teammates. The lineup strength feature constructed here is simpler than those methods. It is a shrunk per minute plus or minus summed across the five players on the floor. That is a scope choice for the model and it is described accurately rather than disguised as something more intense.

A second, concerns how much confidence any win probability estimate deserves. Brill, Yurko and Wyner (2025) show that fourth-down decision recommendations built on win probability models carry far more uncertainty than the point estimates suggest, which is the argument this paper takes seriously in reporting an interval rather than a single number for its key comparison.

The third concerns clustered data. It is well established that event-level observations drawn from the same game are not independent, and that treating them as independent leads to confidence intervals that are overly narrow. Less frequently discussed is the same issue arising during feature construction rather than during statistical inference. Section 7 of this paper presents the positive result of addressing that concern directly.

3. Data

3.1 Source and Scope

All data were pulled through the NBA API package, using the PlayByPlayV3 and BoxScoreTraditionalV3 endpoints. The scope is all 636 Boston Celtics regular season games from 2016-17 through 2023-24, comprising 308,975 individual play-by-play events. Both endpoints returned complete data for all 636 games, with zero failures and zero empty payloads.

The target variable was validated against an external reference. Boston's record over the period came back as 413-223, which matches the published season records summed over exactly 636 games. The pipeline never had access to that reference, so the agreement is genuine evidence rather than a circular check. Every individual game result and point total was also checked against the boxscore totals, and all 636 agreed. The target is verified on the full dataset, not on a sample.

3.2 Defects Found in the Historical Feed

The following defects were found by testing rather than assumed, and none of them were silently repaired. This section is a contribution in its own right, since any study that uses this feed without checking for these defects will produce results that look reasonable, but are incorrect.

Table 1. Defects found in the NBA historical play-by-play feed.

Defect	Scale	Consequence if unhandled
The 2016-17 season encodes "no score" as the string "0" rather than as an empty value	27,462 events	74 percent of margins wrong in one game tested; a full season silently corrupted
Trailing events report a stale score	isolated	non-monotone score sequences
Action Number is not unique within a game	521 events in one game	sorting on it reorders events and breaks the clock
Substitutions between periods are frequently unlogged	widespread	36 percent of lineups corrupted in early testing
Substitutions name only the outgoing player by identifier	all 29,186	the incoming player must be resolved from free text
Team plus minus is unreliable	10 of 636 games	wrong margins if used
The position field is unusable before 2017-18	164 team games	wrong starters for a full season

The first of these defects in the data deserves emphasis. “The 2016-17 season encodes...” was invisible in aggregate. Final scores reconciled correctly and margins fell within believable ranges. Only a check run across the full dataset found this defect. That is the argument for validating everything rather than sampling.

A further measurement is worth reporting on its own. Across all 636 games, 31.2 percent of logged substitution events are internally inconsistent, with a per game range from 0 to 60.4 percent. The pipeline handles these defensively rather than allowing them to corrupt the lineup state.

3.3 Resolving Substitution Names

Rebuilding the five players on the floor requires matching names, and the naming problems in this feed are more interesting than they first appear. Diacritics differ between fields, so the same player appears as K. Porzingis with diacritics in one place and Porzingis without them in another. Suffixes are part of the surname and are load bearing, because Boston carried both Grant Williams and Robert Williams III at the same time. Teammates who share a surname are given a variable length first name prefix, lengthened only as far as is needed to separate them, which is why the identical twins Marcus and Markieff Morris appear as Marc and Mark Morris. And name fields are current while description text is historical, so Enes Kanter appears as Freedom in the roster and as Kanter in the description text under the same identifier.

All 29,186 substitutions were resolved and none were presumed. The resolution paths were 28,605 by roster surname, 313 by initial plus surname, 205 by game local alias, 62 by first name prefix, and 1 by cross game alias.

3.4 Lineup Reconstruction and its Validation

The reconstructed lineups were checked against an independent source. Derived minutes match boxscore minutes to within 0.008 minutes across 13,546 player games, with a mean absolute difference of 0.0013 minutes. Both lineups contain exactly five players at all 308,975 events.

This is the load bearing check in the data section. The boxscore knows nothing about how the lineups were rebuilt, so agreement between the two is independent evidence that the reconstruction is correct. One caveat is stated honestly: for 5 of the 636 games the assignment was chosen by minutes reconciliation, so for those games the minutes check is not fully independent. For the other 631 games it is.

4. Features and Experimental Design

4.1 The Features

The CLWP model uses thirteen features, all of which describe the state of the game at the moment of the event being predicted.

- `celtics_margin`, the score margin from Boston's perspective rather than the raw home and away scores
- `seconds_remaining_period` and `seconds_remaining_game`
- `seconds_elapsed_game`
- `period` and `is_overtime`
- `celtics_is_home`
- `celtics_has_possession`
- `momentum_120s` and `momentum_300s`, the net scoring over the previous two and five minutes
- `is_clutch`
- `margin_per_minute_remaining`
- `possession_number`

Possession is derived by a state machine that walks through made and missed shots. It also includes rebounds, including team rebounds, the offensive or defensive distinction, turnovers, and multi shot free throw trips. Across all 636 games Boston is recorded as holding the ball on 49.4 percent of events, which confirms there is no directional bias in the derivation.

4.2 The Model Tiers

Four tiers were fitted, each adding information to the one before it.

- **Tier 1**, pregame only. Home court advantage and nothing else.
- **Tier 2**, the generic model based on ESPN. Score margin and time remaining only.
- **Tier 3**, the CLWP model. The thirteen game state features listed above.
- **Tier 4**, the CLWP model plus lineup strength.

The classifier is gradient boosted trees, XGBoost (Chen and Guestrin, 2016), with a fixed random seed of 20261730.

4.3 Validation Design

Every result in this paper is out of fold. The model was trained eight separate times using the leave one season out cross validation. On each of the eight folds, one complete season was withheld and the model was then scored only on that withheld season. No prediction reported here comes from a model that had seen the game it is predicting, or any other game from that season.

Two metrics carry most of the argument and both are defined here. The Brier score (Brier, 1950) measures the accuracy of probability predictions, not only if a prediction was correct or wrong. A lower Brier score is better, with zero being perfect. The Brier score rewards being confidently correct and penalizes being confidently wrong. In basketball terms, if the model indicates Boston has a 70 percent chance to win, then Boston should win about 7 out of 10 similar games. If a model consistently predicts 99 percent and Boston lose more often than that implies, the Brier score becomes much higher, and worse, as a result of incorrect overconfidence. Brier skill expresses the same idea relative to a model that never learns anything, and is the best and most accurate way to read a Brier score on its own.

Expected calibration error (ECE), computed using the binned approach described by Guo et al. (2017), measures a different property. Instead of evaluating if the model ranks outcomes correctly, it asks whether predicted probabilities are well calibrated. For example, among all instances where the model predicted a 70% chance of a Boston win, Boston should actually win about 70% of those games. A model may rank games accurately while consistently producing probabilities that are too confident or too conservative, and ECE is designed to detect this miscalibration.

Differences between models are tested with a cluster bootstrap (Efron and Tibshirani, 1993) that resamples games rather than events, with n equal to 636. This is the correct unit. Events inside a single game share an outcome and are highly dependent, so resampling 308,975 correlated rows as though they were independent would produce an interval far too narrow to be meaningful. The bootstrap is paired, meaning both models are scored on the same resampled games. No multiplicity correction was applied across the comparisons reported, which is noted again in the limitations.

5. Results

5.1 The Tier Comparison

Table 2. Out of fold performance by tier, pooled over 308,975 events and 636 games, against a base rate of 0.6447.

Model	AUC	Brier	Skill	Log loss	ECE
Tier 1: pregame only	0.5315	0.2258	1.4%	0.6436	0.0267
Tier 2: generic model based on ESPN	0.8119	0.1641	28.4%	0.4917	0.0072
Tier 3: the CLWP model	0.8155	0.1630	28.8%	0.4871	0.0140
Tier 4: CLWP plus lineup strength	0.7768	0.1823	20.4%	0.5429	0.0606

This is read as steps and the largest gain, by a wide margin, comes from using in-game information. Moving from a pregame prior to the generic model based on ESPN improves the Brier score from 0.2258 to 0.1641 and the AUC from 0.5315 to 0.8119.

The CLWP model then improves on the generic model, moving the Brier score from 0.1641 to 0.1630 and the AUC from 0.8119 to 0.8155. Adding lineup strength makes the model substantially worse, moving the Brier score from 0.1630 to 0.1823.

5.2 The Value of the CLWP Model Against a No-Skill Forecaster

The best way to read a Brier score of 0.1630 is against a forecaster who does not really forecast at all. Boston won 64.5 percent of the games used in this study. A person who says "64.5 percent" at every single moment of every game, and never reacts to anything that happens on the floor, scores a Brier of 0.2290.

The CLWP model updates its prediction continuously, in real time, as the game develops, and scores 0.1630. CLWP therefore reduces the prediction error by 28.8 percent compared with the always 64.5 percent approach. that is, the CLWP model moves a little over a quarter of the way from a static, no-skill prediction toward perfect forecasting.

5.3 The Pooled Comparison Against the Generic Model

The comparison the project was designed to evaluate is between the CLWP model and the generic model and the pooled results indicate that the two are statistically the same.

The CLWP model outperformed the generic model by 0.0011 Brier points. However, when the comparison was repeated across 1,000 bootstrap resamples of the same 636 games, the estimated difference ranged from -0.0029 to +0.0048. Because this interval includes zero, the observed difference is not statistically significant. Accordingly, the pooled results indicate that the two models are statistically indistinguishable.

This conclusion is stated explicitly rather than being hidden by the point estimate alone. Considered by itself, the 0.6% relative improvement would suggest a modest advantage and appear to support the preregistered hypothesis. However, that conclusion does not hold once uncertainty is quantified using an interval computed on the appropriate unit of analysis.

Table 3. Pre-registered comparisons, cluster bootstrap over games, $n = 636$. A positive difference means the second model listed is better.

Comparison	Difference	95% interval	Distinguishable ?
Generic model vs CLWP model	+0.0011	[-0.0029, +0.0048]	no
CLWP vs CLWP plus lineup strength	-0.0193	[-0.0252, -0.0133]	yes
CLWP vs lineup difference only	-0.0112	[-0.0167, -0.0056]	yes
CLWP vs heavy shrinkage lineup	-0.0148	[-0.0203, -0.0094]	yes
CLWP vs time-interacted lineup	-0.0113	[-0.0168, -0.0057]	yes
CLWP vs opponent context tier	-0.0489	[-0.0663, -0.0319]	yes
CLWP vs opponent point differential	-0.0368	[-0.0509, -0.0233]	yes
CLWP vs opponent recent form	-0.0282	[-0.0423, -0.0150]	yes

5.4 Lineup Strength

Adding lineup strength to the CLWP model substantially worsens performance, increasing the Brier score by 0.0193 points, with an interval of [-0.0252, -0.0133]. Three alternative parameterizations were specified in advance rather than selected after observing the initial result, and all three also produced

worse performance. This provides a decisive negative answer to the second research question, and the effect is clearly not marginal.

5.5 Opponent Quality

Every specification of opponent quality tested made the model worse, and worse by a wide margin. This shows a robust negative result across four independent attempts.

The framing of this result requires careful interpretation. The appropriate conclusion is not that opponents do not matter, as that would overstate what was tested. Rather, the evidence indicates that static, season-long measures of opponent strength did not provide reliable incremental predictive value beyond the game-state features already included in the CLWP model. Opponent quality clearly matters in basketball; however, the most likely explanation is that the live game state—particularly score margin and momentum-related features—already captures much of the information that opponent strength would provide. As a result, these season-average opponent features appear to be largely redundant and introduce additional noise rather than meaningful predictive inputs.

6. Where the CLWP Model Adds Value

The pooled comparison in Section 5.3 is not the whole picture, and the part it conceals is the part that matters most for a live display.

Table 4. Skill and expected calibration error by game phase, generic model against the CLWP model.

Phase	Events	Generic skill	CLWP skill	Generic ECE	CLWP ECE
first 6 min	33,553	-0.3%	1.4%	0.0474	0.0272
period 1	73,901	2.4%	4.0%	0.0428	0.0172
period 2	77,351	14.9%	15.2%	0.0250	0.0208
period 3	75,451	33.5%	33.0%	0.0276	0.0236
period 4	79,567	59.9%	60.5%	0.0353	0.0106
overtime	2,705	35.7%	30.5%	0.0759	0.0839
clutch	15,102	33.3%	32.0%	0.0496	0.0424
last 2 min	19,337	75.4%	75.0%	0.0139	0.0211

The CLWP model is better calibrated than the generic model based in seven of the eight phases. The single exception is overtime, which is also by far the smallest phase sample of 2,705 events.

The early game difference between the models is the clearest and the most useful. In the first six minutes the generic model has negative skill, at -0.3 percent, which means it performs worse than simply predicting Boston's base rate. The CLWP model manages positive skill at 1.4 percent over the same events, and its expected calibration error is about 43 percent lower. When little has happened on the court, knowing the identity of the teams provides meaningful predictive value, which is precisely where team-specific information would be expected to matter most. Once the score margin has developed, however, the game state becomes the dominant source of information, and the two models converge.

One apparent contradiction should be addressed directly. The pooled expected calibration error favors the generic model, at 0.0072 against 0.0140, even though the CLWP model is better calibrated inside seven of

the eight phases. This is not an inconsistency in the results. Calibration error does not decompose across strata. Pooling mixes phases whose probability distributions are very different, and a model can be closer to correct within every phase while a pooled average of the same predictions is not. For an application that states a probability at a particular moment, calibration within the current phase is the property that matters and on that criterion where the CLWP model is the better of the two.

7. Two Findings About Event-Level Models

7.1 A Cardinality Artifact

The opponent tiers did not simply fail. They failed in a way that is worth reporting as it points at a general problem with event level sports models.

When a high-cardinality identifier is added as a feature, a gradient-boosted tree can use it to memorize individual games rather than learn patterns that generalize. To test whether this was occurring, the opponent identifier was replaced with a randomly generated identifier of the same cardinality that contained no predictive information. The random control reproduced 85% of the performance degradation caused by the actual feature: the opponent-tier model produced a Brier score of 0.2119, while the random identifier model produced a Brier score of 0.2044, compared with 0.1630 for the CLWP model.

This finding has important implications for interpreting these models. Standard feature-importance analyses can be misleading when a feature is being memorized rather than meaningfully used, because a memorized feature may appear highly influential while providing no improvement in out-of-sample performance. For this reason, no feature-importance plots are presented in this paper.

The same effect is visible in the deployed model rather than only in the research pipeline. The dashboard's what-if control, which re-predicts a real moment with the score margin changed, uses the deployment model fitted on all eight seasons and is therefore in-sample. At large deficits it returns memorized outcomes rather than forecasts. One example is an event with Boston down 25 points in the third quarter of a game they went on to win returns 33.2 percent from the out of fold model and 77.2 percent from the deployment model. Only 46 of the 636 games ever reached a 20 point deficit in the third quarter, and Boston won 3 of them so the deployment model is not forecasting in that region it is identifying the specific game. The dashboard caps the control and labels the panel as in-sample for this reason. No number reported in this paper is affected, because every paper figure is out of fold.

7.2 What The Model Cannot Do Before Tip-Off

At the moment before tip-off, every in-game feature is empty. The score is 0-0, no time has elapsed, there is no momentum and there is no possession. The model is being asked a question its inputs cannot answer.

What it does in that situation is instructive. Across all 636 games the CLWP model emits exactly two values before tip-off: 58.0 percent when Boston are away and 71.0 percent when Boston are at home. It never once starts Boston below 50 percent, in any of the 636 games, regardless of opponent. In the 58 games where Boston was both away and considered the weaker team, the model opened at an average of 58.5 percent where Boston went on to win 37.9 percent of the games, an error of about 20 percentage points.

A decaying pregame prior was developed and evaluated as a potential improvement. It materially improves predictions at tip-off, reducing Brier score by 0.0089 points with an interval of $[+0.0021, +0.0155]$. However, when evaluated across the entire game, the improvement is only 0.0021 Brier points,

with an interval of $[-0.0003, +0.0044]$, which does not meet the preregistered threshold for adoption. Accordingly, the prior is reported as an evaluated modification but not incorporated into the final model, following the same standard applied to the lineup tier and opponent-strength specifications. Selectively adding it to the CLWP model alone would have biased the comparison by improving one model after the fact, effectively manufacturing a positive answer to the study's central question.

8. Limitations

The following limitations are stated in the paper rather than left for the reader to find.

- The pooled AUC of 0.8155 is flattered by fourth quarter events, where a large lead genuinely does decide the game and any model looks capable. The early game and close game rows in Table 4 are the informative ones.
- A forward chaining validation, which trains only on earlier seasons, scores the CLWP model 0.0025 Brier worse than leave one season out, with an interval of $[-0.0047, -0.0005]$. The figures reported throughout Section 5 should therefore be read as modestly optimistic relative to what the model would achieve forecasting a genuinely unseen future season. That gap is confounded, because forward chaining both withholds the future and trains on less data, and the confound is stated rather than resolved.
- No multiplicity correction was applied across the eight or more intervals reported. With that many comparisons, some caution about any single borderline interval is warranted.
- The lineup strength construction is deliberately simple and is not adjusted plus minus. A more sophisticated player valuation might behave differently, and this study does not test that.
- The cardinality permutation check used only three permutations, which is enough to establish the direction of the effect and not enough to place a tight interval on its size.
- The study covers one team. The conclusions about team-specific information may not transfer to a team with a very different competitive profile over the same period.

9. Conclusion

The CLWP model performs well. In out-of-fold evaluation across 308,975 events and 636 games, it achieves a Brier score of 0.1630 and closes 28.8% of the gap between a static no-skill forecast and a perfect forecast. Importantly, every prediction is generated from a model that was trained without exposure to that season.

Relative to the ESPN-based generic model, the pooled result is a statistical tie. The 0.0011 Brier-point difference has an interval of $[-0.0029, +0.0048]$, which includes zero. Reporting this result directly provides the appropriate answer to the first research question.

The phase-by-phase results add important context to this conclusion. The CLWP model is better calibrated in seven of eight game phases, with its clearest advantage occurring in the opening minutes, where the generic model performs worse than a no-skill baseline. Team-specific information is most valuable when little else is available and becomes less important once the evolving game state has revealed the outcome trajectory. For a live prediction system that must provide defensible probabilities at every moment—including the earliest moments of the game—this distinction is critical.

The answers to the second and third research questions are negative and decisive. Neither lineup strength nor any of the four tested specifications of opponent quality improved the model; instead, each specification degraded performance.

Two broader lessons extend beyond this project. First, high-cardinality identifiers in event-level models can be memorized rather than learned, making conventional feature-importance interpretations unreliable precisely in situations where practitioners may be most inclined to trust them. Second, a model with no game-specific information at the moment it is first asked to make a prediction will necessarily fall back on the average patterns in its training data—and it will express that uncertainty no differently from predictions made later in the game when meaningful information is available.

10. Reproducibility and The Dashboard

The pipeline is reproducible end to end in Python. Raw API responses are immutable and every transformation downstream of them is scripted. Random seeds are fixed. Failures raise rather than returning empty, because an empty frame is indistinguishable from a game with no events. The audit that vouches for the game index is a separate program from the one that builds it, and the test suite deliberately corrupts synthetic data in eleven ways to prove the audit catches the problems it claims to catch.

The dashboard is a browser application that replays any of the 636 games and shows the CLWP model's probability after every play. The trained gradient boosted tree is exported to JSON and evaluated in JavaScript, which was verified against the Python model on all 308,975 rows to a maximum absolute difference of $4.837e-07$. Every probability shown on the game timeline is out of fold. The one in-sample surface, the what-if control described in Section 7.1, is labeled as such in the interface.

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